# Spec → Code: RID171

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## 1. Spec

**REPORT ID**

RID171

**EDC SYSTEM / LANGUAGE**

veeva / CQL

**DESCRIPTION**

ADA data missing in InForm or Vendor transfer file - SCI eCRF AND CENTRAL LAB/BA VENDOR RECONCILIATION.
The report will output when there is a SCI record in eCRF with no matching record in the external vendor transfer file.
The report will also output when there is a ADA record in the external vendor transfer file that has no matching record in InForm.

**COLUMN HEADING**

Tab 1: Inform Data with BA Vendor Data
1. Country
2. Site
3. Screening Number
4. Subject Number
5. InForm Visit Name
6. InForm Analyte
7. InForm Schedule Time
8. InForm Spec ID
9. Inform Colection Date
10. Inform Colection Time
11. BA NOMDAY
12. BA ANALYTE
13. BA NOMHOUR
14. BA Specimen ID
15. BA Colection Date
16. BA Colection Time
17. Error Message

Tab 2: Inform Data with Central Vendor Data
1. Country
2. Site
3. Screening Number
4. Subject Number
5. InForm Visit Name
6. InForm Analyte
7. InForm Schedule Time
8. InForm Spec ID
9. Inform Colection Date
10. Inform Colection Time
11. Central NOMDAY
12. Central ANALYTE
13. Central NOMHOUR
14. Central Specimen ID
15. Central Colection Date
16. Central Colection Time
17. Error Message

**LOGIC**

Scenario 1

1A
Output is displayed when a combination of screening number and Specimen ID entered in InForm cannot be found in vendor data. Output error message "Missing in Vendor"

1B
Output is displayed when a combination of screening number and Specimen ID loaded from vendor cannot be found in InForm data. Output error message "Missing in Inform"

Notes:
Note1. Logic takes the SCI record into consideration only when "Reason Specimen Not Collected and/or Not Sent" in that record is blank.
Note2. Please do not consider deleted records from SCI forms in this logic.
Note3. Report is NOT Case Sensitive between Specimen ID entered in InForm and Specimen ID loaded from vendor.
Note4 : DOV should be initialized, and visit should not be erroneous for the record to be considered in the logic.
Note5. SITEID is always a 4-digit number, SCRNID is a 9-digit number and SUBJID is a 6-digit number if any of them have a smaller number of digits please complete with zeros at the left to make the comparison, i.e. SITEID: 100 should be 0100 to match with Site from InForm
Note6. In case of scenario "Missing in Inform", the Site and Screening Number will be displayed from vendor. The Subject number will be a combination of Site and Subjectid: For example, BA/Central.Siteid + "--" + BA/Central.Subjid

**REPORT SCENARIOS**

Scenario 1

1A Output = Yes
Output is displayed when a combination of screening number and Specimen ID entered in InForm cannot be found in vendor data. Output error message "Missing in Vendor"

1A Output = No
Output is not displayed when a combination of screening number and Specimen ID entered in InForm can be found in vendor data.

1B Output = Yes
Output is displayed when a combination of screening number and Specimen ID loaded from vendor cannot be found in InForm data. Output error message "Missing in InForm"

1B Output = No
Output is not displayed when a combination of screening number and Specimen ID loaded from vendor can be found in InForm data.

**SORTING**

default

## 2. Logic Understanding

This spec implements a two-directional, per-vendor reconciliation between InForm SCI eCRF specimen records and two separate external vendor transfer files (BA Vendor on Tab 1, Central Vendor on Tab 2). After filtering InForm records to exclude deleted, 'not collected/not sent', DOV-uninitialized, or erroneous-visit records, it matches on the combination of Screening Number + Specimen ID (case-insensitive, with zero-padded IDs) in both directions: InForm records without a vendor match flag 'Missing in Vendor', and vendor records without an InForm match flag 'Missing in Inform', with the latter scenario sourcing Site/Screening/Subject Number fields from the vendor data per Note6.
**1.** This report reconciles specimen collection records entered in InForm (via the SCI eCRF form) against two separate external vendor transfer files: a BA Vendor file (output on Tab 1) and a Central Lab Vendor file (output on Tab 2). Each tab runs the same two-directional check independently against its respective vendor file. *(inferred)*

**2.** Before any matching occurs, filter the InForm SCI records to only those eligible for the logic: exclude deleted SCI records (Note2), exclude records where 'Reason Specimen Not Collected and/or Not Sent' is not blank (Note1), and exclude records where DOV is not initialized or the visit is marked erroneous (Note4).

**3.** For matching purposes, standardize identifiers: SITEID must be zero-padded to 4 digits, SCRNID zero-padded to 9 digits, and SUBJID zero-padded to 6 digits (Note5), so that comparisons between InForm and vendor values align correctly even if one side has fewer leading digits.

**4.** Specimen ID comparisons between InForm and vendor data are case-insensitive (Note3), so differing letter case alone should not cause a mismatch.

**5.** Scenario 1A: For each eligible InForm SCI record, form the combination of Screening Number + Specimen ID and check whether that combination exists anywhere in the vendor data (BA vendor for Tab 1, Central vendor for Tab 2). If no matching combination is found in the vendor data, output that InForm record's row with Error Message = 'Missing in Vendor'.

**6.** Scenario 1A Output=No condition: if the InForm record's Screening Number + Specimen ID combination IS found in the vendor data, it is not output - no discrepancy exists.

**7.** Scenario 1B: For each record loaded from the vendor data (BA or Central, respectively), form the combination of Screening Number + Specimen ID and check whether that combination exists among eligible InForm SCI records. If no matching combination is found in InForm, output that vendor record's row with Error Message = 'Missing in Inform'.

**8.** Scenario 1B Output=No condition: if the vendor record's Screening Number + Specimen ID combination IS found in InForm data, it is not output - no discrepancy exists.

**9.** When outputting a 'Missing in InForm' row (Scenario 1B), since there is no corresponding InForm record to source Site/Screening Number/Subject Number from, these fields are populated from the vendor data instead: Site and Screening Number come directly from vendor, and Subject Number is constructed as vendor's Siteid + '--' + vendor's Subjid (Note6).

**10.** When outputting a 'Missing in Vendor' row (Scenario 1A), by contrast, the identifying fields (Country, Site, Screening Number, Subject Number) are presumably populated from the InForm SCI record itself, since that's the record that exists and is missing its vendor counterpart. *(inferred)*

**11.** Each output row also includes the full set of available data fields from whichever side has data (InForm columns like Visit Name, Analyte, Schedule Time, Spec ID, Collection Date/Time always populated since it's the source of comparison; and the corresponding BA/Central columns populated only when a partial match or the vendor side has data), though the spec's Note6 implies the missing side's fields would simply be blank when no matching record exists on that side. *(inferred)*

## 3. Spec Review

Readiness: Needs revision — The spec has a solid narrative description and helpful normalization notes, but the Logic section is written as a single generic scenario that doesn't reflect the two distinct vendor sources (BA vs Central) implied by the Description and the two-tab Column Heading structure, leaving matching/output-mapping rules for Tab 1 vs Tab 2 undefined. Several vendor-specific columns (NOMDAY, NOMHOUR, dates/times, Country) are never tied to Logic, so a generator would have to guess at population rules.

**BLOCKING (consistency):** Description and Column Headings describe two distinct vendor sources (BA VENDOR and CENTRAL LAB/Vendor), producing two tabs with different data (BA Specimen ID/NOMDAY/NOMHOUR vs Central Specimen ID/NOMDAY/NOMHOUR). However, the Logic section only defines a single, generic 'Scenario 1' (1A/1B) that references 'vendor data' as one undifferentiated source, and never distinguishes BA vendor data from Central vendor data, nor states how a record is routed to Tab 1 vs Tab 2. → Add explicit logic branches (e.g., 1A-BA, 1B-BA for Tab 1 and 1A-Central, 1B-Central for Tab 2) or a rule stating how InForm/SCI records are matched against each vendor source independently, and how output rows are assigned to each tab.

**BLOCKING (completeness):** Column Headings include vendor-specific fields such as 'BA NOMDAY', 'BA NOMHOUR', 'Central NOMDAY', 'Central NOMHOUR' but the Logic (1A/1B) never mentions NOMDAY or NOMHOUR as part of the matching key or as populated output fields -- only 'screening number and Specimen ID' are used for matching, and Colection Date/Time are not mentioned either. → State explicitly which columns (NOMDAY, NOMHOUR, Collection Date, Collection Time, Visit Name, Schedule Time, Analyte) are sourced directly from the matched record (or left blank when missing) versus used in matching logic, for both the 'Missing in Vendor' and 'Missing in InForm' scenarios.

**BLOCKING (completeness):** Note6 explains how Site/Screening Number/Subject Number are populated for the 'Missing in InForm' scenario (from vendor: Site, Screening Number, and Site+'--'+Subjid), but there is no equivalent rule for the 'Missing in Vendor' scenario describing how the InForm-side Country/Site/Screening Number/Subject Number map to columns, nor how Country is derived at all (Country does not appear as a Logic input for either scenario). → Add a note specifying the source of the Country column for both scenarios, and confirm that for 'Missing in Vendor' the Site/Screening/Subject Number are taken directly from InForm's SCI record fields without transformation.

**WARNING (ambiguity):** Note4 states 'DOV should be initialized, and visit should not be erroneous for the record to be considered in the logic' but does not define which record(s) this applies to (SCI/InForm only, or also vendor records), nor what 'DOV initialized' or 'visit erroneous' means operationally (e.g., a specific status flag or field). → Clarify which form(s)/record types Note4 applies to and define 'DOV initialized' and 'visit erroneous' in terms of specific field names/values (e.g., a QueryStatus or VisitStatus flag).

**WARNING (completeness):** The report is a reconciliation between InForm/SCI and two separate vendor feeds, but the spec never addresses what happens if a Specimen ID/Screening Number pair appears in both InForm and one vendor source but not the other, or if duplicate Specimen IDs exist within a single source -- an edge case likely to occur in reconciliation reports. → Add a note describing behavior when duplicate Specimen ID/Screening Number combinations exist within InForm or vendor data, and how such duplicates are matched/reported.

**WARNING (testability):** Report Scenarios (1A Output=Yes/No, 1B Output=Yes/No) restate the Logic verbatim without adding independently verifiable conditions, and do not map each scenario to which Tab (1 or 2) or which vendor source it exercises, making it hard to construct distinct test cases for BA vs Central reconciliation. → Expand Report Scenarios to include separate test cases for BA-vendor matching and Central-vendor matching (e.g., 1A-BA Yes/No, 1A-Central Yes/No), so each tab's logic is independently testable.

**INFO (naming):** Column headings use inconsistent/misspelled terms such as 'Colection Date/Time' (missing a 'l') and mix of 'InForm'/'Inform'/'INFORM' capitalization across the Description, Column Headings, and Notes (e.g., 'Inform Colection Date' vs 'InForm Visit Name' vs Note6 'BA/Central.Siteid'). → Standardize spelling to 'Collection Date/Time' and consistent capitalization of 'InForm' throughout the spec.

**INFO (ambiguity):** Note5 gives an example only for SITEID left-padding ('100 should be 0100') but does not explicitly confirm the same left-padding rule applies to SCRNID and SUBJID matching keys used in 1A/1B, though it's implied by the sentence structure. → Explicitly state that the same zero-padding normalization applies to Screening Number and Subject ID comparisons, not just Site.

## 4. Generated Code

CQL — Guardrails: passed

```
SELECT
  d1.SiteCountry AS `Country`,
  d1.SiteNumber AS `Site`,
  d1.ScreenNum AS `Screening Number`,
  d1.SubjNum AS `Subject Number`,
  d1.VisitName AS `InForm Visit Name`,
  d1.Analyte AS `InForm Analyte`,
  d1.SchedTime AS `InForm Schedule Time`,
  d1.SpecID AS `InForm Spec ID`,
  d1.CollDate AS `Inform Colection Date`,
  d1.CollTime AS `Inform Colection Time`,
  CAST(NULL AS STRING) AS `BA NOMDAY`,
  CAST(NULL AS STRING) AS `BA ANALYTE`,
  CAST(NULL AS STRING) AS `BA NOMHOUR`,
  CAST(NULL AS STRING) AS `BA Specimen ID`,
  CAST(NULL AS STRING) AS `BA Colection Date`,
  CAST(NULL AS STRING) AS `BA Colection Time`,
  'Missing in Vendor' AS `Error Message`
FROM
  (
    SELECT
      `@HDR`.`Site`.`Country` AS SiteCountry,
      `@HDR`.`Site`.`Number` AS SiteNumber,
      LPAD(CAST(SCICEBESPIDIT AS STRING), 9, '0') AS ScreenNum,
      `@HDR`.`Subject`.`Name` AS SubjNum,
      `@HDR`.`Event`.`Name` AS VisitName,
      SCICECATSOIT AS Analyte,
      SCICETPTIT AS SchedTime,
      SCICEREFIDIT AS SpecID,
      SCICESTDTCSOIT AS CollDate,
      SCICESTDTCSOTMIT AS CollTime
    FROM `EDC`.`SCIFM`
    WHERE `@Form`.`Status` = 'submitted__v'
      AND (DELETEDFM = 'N' OR DELETEDFM IS NULL)
      AND (SCIACERSNNCLNSIT IS NULL OR SCIACERSNNCLNSIT = '')
      AND `@HDR`.`Event`.`Date` IS NOT NULL
      AND (`@HDR`.`Event`.`Erroneous` = 'N' OR `@HDR`.`Event`.`Erroneous` IS NULL)
  ) d1
WHERE NOT EXISTS (
  SELECT 1
  FROM `EDC`.`frmCLAB` v
  WHERE UPPER(LPAD(CAST(v.itmCLABSeqNum AS STRING), 9, '0')) = UPPER(d1.ScreenNum)
)

UNION ALL

SELECT
  d2.SiteCountry AS `Country`,
  d2.SiteNumber AS `Site`,
  d2.ScreenNum AS `Screening Number`,
  d2.SubjNum AS `Subject Number`,
  d2.VisitName AS `InForm Visit Name`,
  d2.Analyte AS `InForm Analyte`,
  d2.SchedTime AS `InForm Schedule Time`,
  d2.SpecID AS `InForm Spec ID`,
  d2.CollDate AS `Inform Colection Date`,
  d2.CollTime AS `Inform Colection Time`,
  CAST(NULL AS STRING) AS `Central NOMDAY`,
  CAST(NULL AS STRING) AS `Central ANALYTE`,
  CAST(NULL AS STRING) AS `Central NOMHOUR`,
  CAST(NULL AS STRING) AS `Central Specimen ID`,
  CAST(NULL AS STRING) AS `Central Colection Date`,
  CAST(NULL AS STRING) AS `Central Colection Time`,
  'Missing in Vendor' AS `Error Message`
FROM
  (
    SELECT
      `@HDR`.`Site`.`Country` AS SiteCountry,
      `@HDR`.`Site`.`Number` AS SiteNumber,
```

```
      LPAD(CAST(SCICEBESPIDIT AS STRING), 9, '0') AS ScreenNum,
      `@HDR`.`Subject`.`Name` AS SubjNum,
      `@HDR`.`Event`.`Name` AS VisitName,
      SCICECATSOIT AS Analyte,
      SCICETPTIT AS SchedTime,
      SCICEREFIDIT AS SpecID,
      SCICESTDTCSOIT AS CollDate,
      SCICESTDTCSOTMIT AS CollTime
    FROM `EDC`.`SCIFM`
    WHERE `@Form`.`Status` = 'submitted__v'
      AND (DELETEDFM = 'N' OR DELETEDFM IS NULL)
      AND (SCICERSNNCLNSIT IS NULL OR SCICERSNNCLNSIT = '')
      AND `@HDR`.`Event`.`Date` IS NOT NULL
      AND (`@HDR`.`Event`.`Erroneous` = 'N' OR `@HDR`.`Event`.`Erroneous` IS NULL)
  ) d2
WHERE NOT EXISTS (
  SELECT 1
  FROM `EDC`.`frmHLAB` v2
  WHERE UPPER(LPAD(CAST(v2.itmHLABSeqNum AS STRING), 9, '0')) = UPPER(d2.ScreenNum)
)
```

This is a best-effort skeleton only: the spec requires reconciling InForm SCI/SCIA (Specimen Collection) forms against two EXTERNAL vendor transfer files (BA vendor and Central vendor), but those vendor transfer files are NOT EDC Forms in the provided study metadata (no BA/Central vendor Form with Siteid/Subjid/Specimen ID/Nomday/Nomhour items exists in the "Known Forms and Items" list) -- I substituted the lab Forms frmCLAB/frmHLAB as placeholders purely to produce a syntactically valid query, but these are almost certainly wrong and must be replaced with the actual vendor-transfer Form/Item names once identified. I mapped InForm fields to SCIFM (SCICEBESPIDIT=screening/subject link, SCICEREFIDIT=Specimen ID, SCICECATSOIT=Analyte, SCICETPTIT=Schedule Time, SCICESTDTCSOIT/SCICESTDTCSOTMIT=Collection Date/Time), used DELETEDFM for Note2, blank SCICERSNNCLNSIT/SCIACERSNNCLNSIT for Note1, and a non-existent `@HDR`.`Event`.`Erroneous` placeholder plus non-null Event Date for Note4 (DOV/erroneous-visit logic could not be confirmed against metadata and needs verification via DESCRIBE FORM). The "Missing in InForm" (1B) direction, zero-padding logic (Note5), and Subject Number concatenation (Note6) could not be reliably built without the real vendor Form schema, so I only implemented the "Missing in Vendor" (1A) half for both tabs as a structural template -- strongly recommend obtaining the actual BA/Central vendor Form and Item names before running this report.